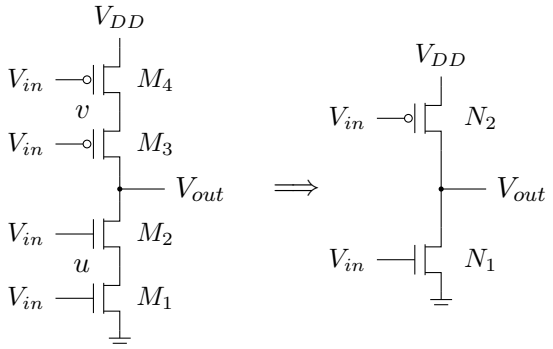
NCU Introduction to VLSI Design, 2018

Homework 1: complete worked solution

1. MOS theory

Let u be the node between the two NMOS devices and v the node between the two PMOS devices. M_1 is nearest ground, M_2 nearest the output; M_4 is nearest V_{DD}, M_3 nearest the output. All gates connect to V_{in}.



(a) M_1 and M_4 cannot be saturated while their corresponding series paths carry nonzero current. For M_1 to saturate, $u \geq V_{in} - V_{tn}$. Then

$$V_{GS,2} = V_{in} - u \leq V_{tn},$$

so M_2 is off (at equality, its ideal overdrive is zero). This contradicts nonzero series current. Thus conducting M_1 must be in the linear region. Likewise, saturation of M_4 requires

$$V_{DD} - v \geq V_{DD} - V_{in} - |V_{tp}| \quad \Longrightarrow \quad v \leq V_{in} + |V_{tp}|.$$

Then $V_{SG,3} = v - V_{in} \leq |V_{tp}|$, so M_3 is off. Conducting M_4 must therefore be linear. M_2 and M_3, the devices adjacent to the output, can saturate. A cutoff device is not called a conducting saturated device here.

(b) Equivalent device sizes. Two equal, common-gate series devices add their channel lengths in the ideal long-channel, body-effect-free model:

$$\boxed{(W/L)_{N_1} = (W/L)_{N_2} = \tfrac{1}{2}(W/L).}$$

For example, keep W and use $2L$, or keep L and use $W/2$. In the linear region the two voltage drops sum and yield half the single-device conductance; the same effective $\beta/2$ follows by eliminating the internal node in the square-law saturation equations. This is the ideal static characteristic equivalence intended by the problem. Internal diffusion capacitances and short-channel/body effects prevent a claim of identical physical transient behavior.

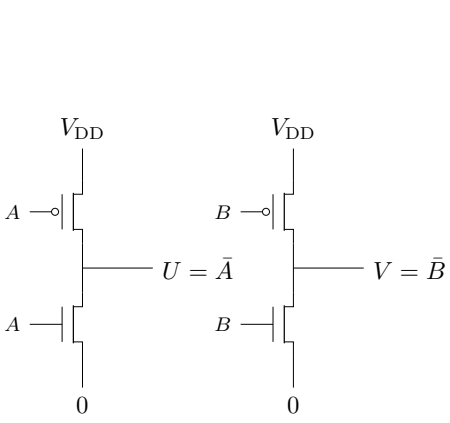
2. Multi-threshold CMOS design

The upper gate in Fig. 2 is XOR, preceded on its A input by an inverter. Consequently

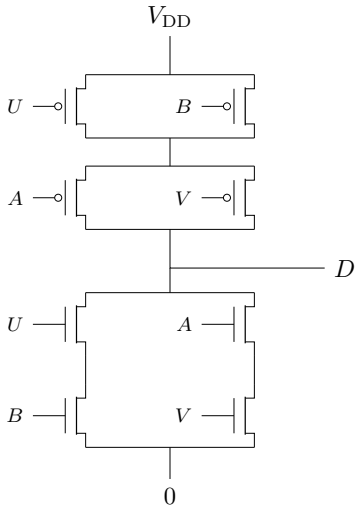
$$U = \bar{A}, \quad V = \bar{B}, \qquad \boxed{D = U \oplus B = AB + \bar{A}\bar{B}}, \qquad \boxed{E = \overline{BC}}.$$

With the stated one-unit low-V_t gate delay, the original paths are $A \to \text{INV} \to \text{XOR} \to D$ (two units), $B \to \text{XOR} \to D$ (one unit), and $B, C \to \text{NAND} \to E$ (one unit). The critical delay is two units. Use low-V_t devices in the A inverter and all devices realizing D. Use high-V_t devices in the four-transistor NAND for E, provided its characterized delay is at most two units. This spends the NAND's one unit of slack without extending the circuit's maximum delay.

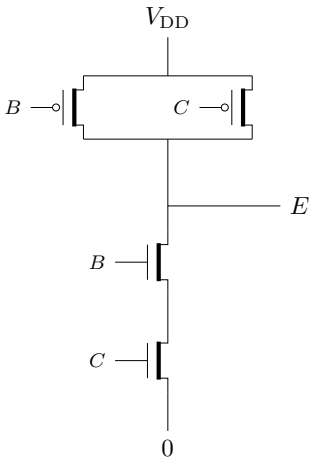
The following complete static CMOS implementation generates both complements locally. Its XOR core uses the pull-down condition $UB + AV$; the pull-up is the exact series/parallel dual. All shown thin channels are low-V_t; the four bold NAND channels are high-V_t. All pMOS bulks connect to V_{DD}, and all nMOS bulks to ground. No body-terminal connections are implicit internal-node ties.



Complement generators: low-V_t



D: low-V_t core



E: high-V_t NAND

The added B complement generator is kept low-V_t: in this specific static implementation the $B \to V \to D$ path also takes two gates. Merely selecting all B-controlled XOR devices as high-V_t would be unjustified because those same series devices can carry a sensitized critical A transition. The prompt gives no high-V_t delay or leakage parameters, so an unconditional numerical timing guarantee or leakage reduction cannot be calculated. If the high-V_t NAND exceeds two units, retain low-V_t devices or resize and characterize it. The standard CIC018 TT model is not a dual-V_t model. `verify_logic.py` checks all eight (A, B, C) assignments.

3. Multiplexer-based design

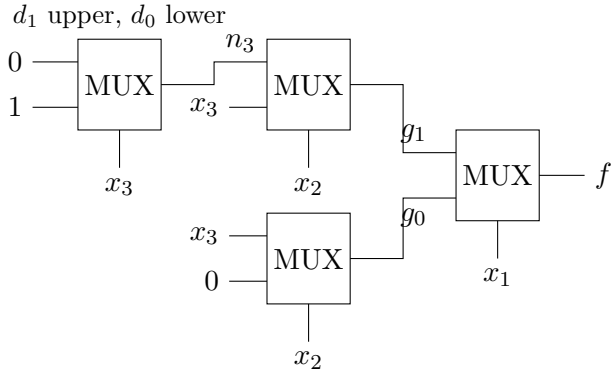
Use $\mathrm{MUX}(s, d_0, d_1) = \bar{s}d_0 + sd_1$. Shannon expansion on x_1 gives

$$f = \bar{x}_1(x_2x_3) + x_1(\bar{x}_2x_3 + x_2\bar{x}_3).$$

The following four 2-to-1 multiplexers implement the function using only true inputs and constants:

$$\begin{aligned} n_3 &= \mathrm{MUX}(x_3, 1, 0) = \bar{x}_3, \\ g_0 &= \mathrm{MUX}(x_2, 0, x_3) = x_2x_3, \\ g_1 &= \mathrm{MUX}(x_2, x_3, n_3) = x_2 \oplus x_3, \\ f &= \mathrm{MUX}(x_1, g_0, g_1). \end{aligned}$$

If $\bar{x}_3$ is already available, only the final three MUXes are required.



x_1	x_2	x_3	f
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

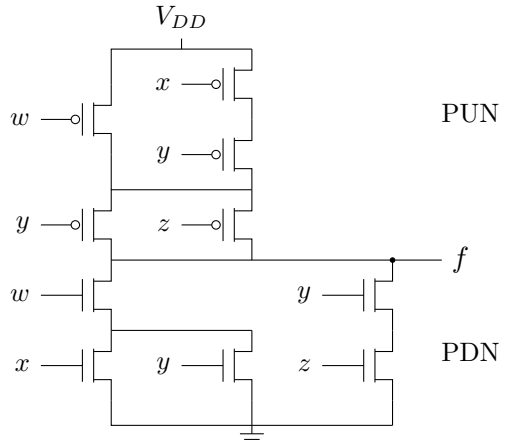
The output is 1 exactly when two of the three inputs are 1.

4(a). Static CMOS: $f = \overline{w(x+y)+yz}$

The pull-down network (PDN) implements $G = w(x+y)+yz$. Its two parallel paths are n_w in series with $(n_x \parallel n_y)$, and n_y in series with n_z. The complementary pull-up network (PUN) is

$$\bar{G} = [\bar{w} + \bar{x}\bar{y}][\bar{y} + \bar{z}].$$

Thus use $p_w \parallel (p_x \text{ series } p_y)$, in series with $p_y \parallel p_z$. All gate labels below are true inputs; PMOS devices turn on when their labeled input is 0. Repeated y labels denote distinct transistors. There are 5 PMOS and 5 NMOS devices.



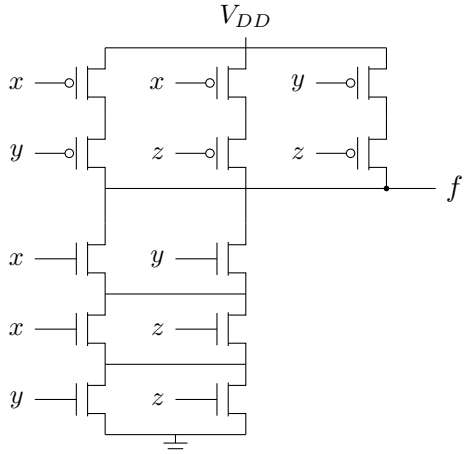
Series connections implement AND in a conducting network and parallel connections implement OR. De Morgan's law makes the networks mutually exclusive for every binary input combination: if $G = 1$, PDN conducts and PUN does not; if $G = 0$, PUN conducts and PDN does not. No input inverter is needed.

4(b). Static CMOS: $f = \bar{x}\bar{y} + \bar{x}\bar{z} + \bar{y}\bar{z}$

Only true literals are supplied, so form the complement of the desired function for the NMOS gates:

$$\bar{f} = (x + y)(x + z)(y + z) = xy + xz + yz.$$

Use three series stages, each containing two parallel NMOS devices, for the PDN. The dual PUN has three parallel branches, each containing two series PMOS devices. It contains 6 PMOS and 6 NMOS devices and needs no external complemented inputs.



This function is 1 for input Hamming weight 0 or 1 and 0 for weight 2 or 3. The product-of-sums network follows the supplied handwritten solution. An alternative PDN is three parallel two-NMOS stacks (xy, xz, yz), with its dual PUN; both implement the same function and use 12 devices.

Source comparison and reproducibility

The authentic two-page Fall 2018 HW1 prompt was inspected visually, including the complete overbar in Q4(a). The supplied three-page handwritten `HW1.pdf` was then read. It agrees on the linear-region devices, half-size static equivalence, Shannon cofactors and complementary CMOS gates. The present solution adds the nonzero-current qualification, the restrictions of the ideal long-channel model, the explicit high-V_t NAND delay bound, and a fourth MUX when the complemented data input is not supplied. The NAND is the only high-V_t block selected without risking a shared critical XOR current path. No characterized dual-threshold process models were supplied, so its timing/leakage condition is theoretical rather than an asserted CIC018 physical measurement.

The independently authored earlier-year derivations were reused only where the actual 2018 printed equations and figures match. The reconstructed problem statements and editable circuit diagrams accompany the independently checked derivations. No missing later homeworks are inferred from these references.

Run `python verify_logic.py`. It verifies 32 original MUX/CMOS cases, eight dual-V_t block logic cases, and 5,050 square-law stack-equivalence points. The logic assertions check complementary rail conduction. The square-law numeric check eliminates the internal stack node and compares both device currents with an equivalent half-strength device. This is not a BSIM timing or leakage simulation. CSV truth tables and a JSON summary are included for reproducibility.